

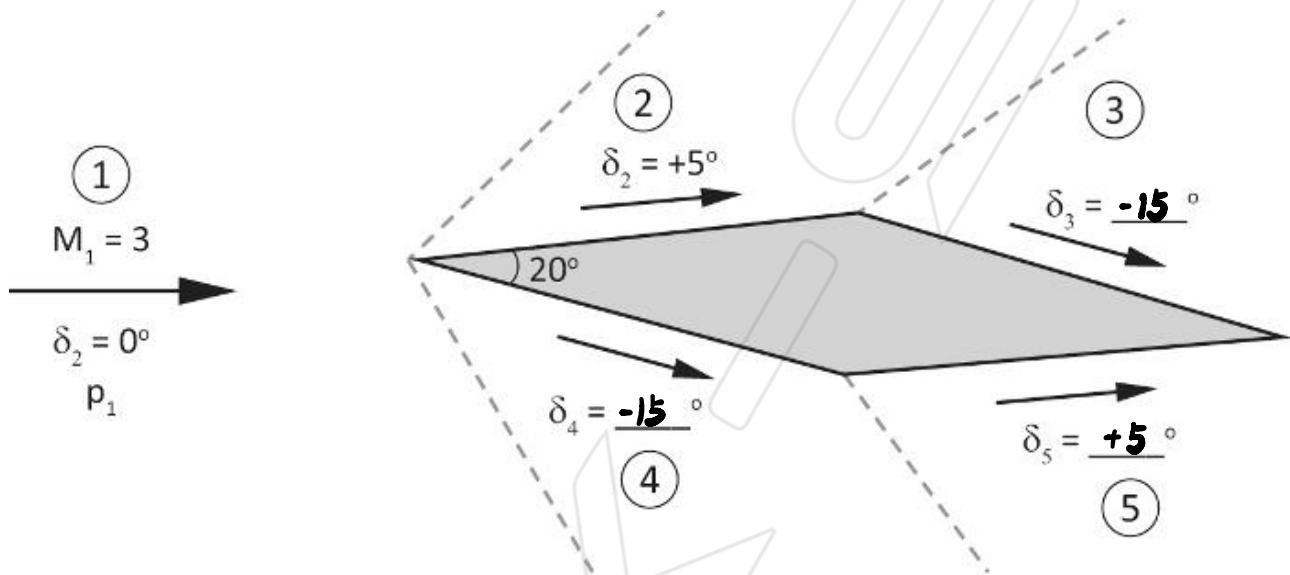
AERO50001 Aerodynamics 2 Problem Set #8

Tutorial 3

8.1 Diamond aerofoil

★★☆ 10-15 mins

The figure below shows a symmetrical diamond shaped aerofoil with a total wedge angle of 20° in a Mach 3 supersonic flow at an incidence of 5°



(a) Determine the flow angle in regions 3, 4, and 5.

(b) Indicate which of the dashed lines are oblique shocks and which are P-M expansion fans

Region 1 to 2:

- Oblique shock ✓
- Expansion fan

Region 1 to 4:

- Oblique shock ✓
- Expansion fan

Region 2 to 3:

- Oblique shock
- Expansion fan ✓

Region 4 to 5

- Oblique shock
- Expansion fan ✓

(c) Use shock-expansion theory to calculate the pressure in region 3 as a fraction of the upstream pressure ($\frac{p_3}{p_1}$).

(d) Repeat part c) using linear theory.

(e) Based on your results from part d), is linear theory appropriate for this problem?

- Yes
- No ✓

(c) from ① to ②, for $\delta_2 = 5^\circ$, $M_1 = 3$

$$\beta = 23.145^\circ, M_2 = 2.75, \frac{p_2}{p_1} = 1.457$$

from ② to ③, for $\delta_2 = 5^\circ$ and $\delta_3 = -15^\circ$

$$\delta_2 + \nu(M_2) = \delta_3 + \nu(M_3)$$

$$\therefore \nu(M_3) = \delta_2 + \nu(M_2) - \delta_3 = 5^\circ + 44.69^\circ + 15^\circ = 64.69^\circ$$

$$\therefore M_3 = 3.918$$

$$\therefore \frac{p_3}{p_2} = \frac{p_3}{p_{03}} \cdot \frac{p_{03}}{p_{02}} \cdot \frac{p_{02}}{p_2} = \frac{p_3}{p_{03}} \frac{p_{02}}{p_2}$$

$$= \frac{1}{136.03} \cdot 25.14 = 0.1848$$

$$\therefore \frac{p_3}{p_1} = \frac{p_3}{p_2} \frac{p_2}{p_1} = 0.2693$$

$$(d) C_{p3} = \frac{2\delta_3}{\sqrt{M_1^2 - 1}} = -0.18512$$

$$\therefore C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2} = \frac{p - p_\infty}{\frac{1}{2} \frac{\rho}{RT} (M_\infty \sqrt{\gamma RT})^2} = \frac{p - p_\infty}{\frac{1}{2} \gamma M_\infty^2} = \frac{2 \left(\frac{p}{p_\infty} - 1 \right)}{\gamma M_\infty^2}$$

$$\therefore \frac{p_3}{p_1} = \left(1 + \frac{C_{p3} \gamma M_1^2}{2} \right) = 0.4616 - 0.166$$

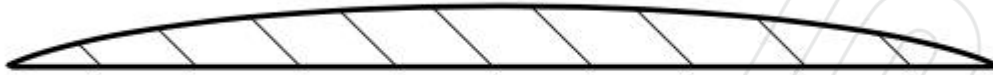
8.2 Linear supersonic theory

★★★ 15-30 mins

The "lenticular" aerofoil sketched below has a thickness/chord ratio of 0.08. It is at an incidence of 4° to a stream at a Mach number of 4. The shape of the upper surface is given by:

$$\frac{y}{c} = 0.32 \frac{x}{c} \left(1 - \frac{x}{c}\right)$$

$$y_c = \frac{y_{upper} + y_{lower}}{2} = 0.16 \frac{x}{c} \left(1 - \frac{x}{c}\right), \quad y_t = y_{upper} - y_c$$



- Use supersonic thin aerofoil theory to evaluate the lift coefficient.
- Use supersonic thin aerofoil theory to evaluate the wave drag coefficient.

$$\begin{aligned} \text{a) } \theta_c &= \frac{dy_c}{dx} = \frac{d}{dx} \left[0.16 \frac{x}{c} \left(1 - \frac{x}{c}\right) \right] = \frac{d}{dx} \left[-0.32 \left(\frac{x}{c}\right)^2 + 0.16 \left(\frac{x}{c}\right) \right] \\ &= -0.64 \left(\frac{x}{c}\right) + 0.16 \end{aligned}$$

$$\theta_t = \frac{dy_t}{dx} = \frac{d}{dx} \left[0.08 \cdot \left[0.32 \frac{x}{c} \left(1 - \frac{x}{c}\right) \right] \right] = 0.03 \theta_c$$

$\frac{y}{c} - y_c$

$$C_L = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} = 0.0721$$

$$\text{d) } C_D = \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\alpha^2 + \int_0^1 (\theta_c^2 + \theta_t^2) d\left(\frac{x}{c}\right) \right)$$

$$= \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\alpha^2 + \int_0^1 (1 + 0.0064) \theta_c^2 d\left(\frac{x}{c}\right) \right)$$

$$= \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\alpha^2 + 0.0064 \int_0^1 \left[-0.64 \left(\frac{x}{c}\right) + 0.16 \right]^2 d\left(\frac{x}{c}\right) \right)$$

$$= \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\alpha^2 + 0.0064 \left[-\frac{1}{0.96} \left[-0.64 \left(\frac{x}{c}\right) + 0.16 \right]^3 \right]_0^1 \right)$$

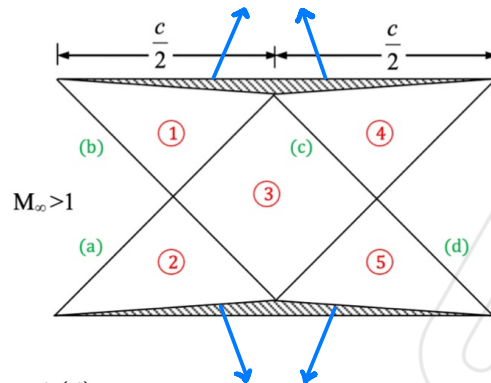
$$= \frac{4}{\sqrt{M_1^2 - 1}} (\alpha^2 + \overset{2}{0.064}) (-\frac{1}{\underset{0.96}{0.4}}) [(-\overset{0.16}{0.3})^3 - (\overset{0.16}{0.3})^3]$$

$$= 0.04\overset{0.51}{051} 2266$$

8.3 Busemann Biplane

★★★ 15-30 mins

A "Busemann Biplane" consists of two identical thin aerofoils arranged as shown in the figure below, each with a deflection angle θ_w .



Note: waves labeled (a), (b), (c) and (d).

regions labels 1, 2, 3, 4 and 5.

∞ refers to free stream.

- Considering a deflection of θ_w , use linear supersonic theory to find the pressure coefficients in regions 1, 2, 3, 4, and 5. Express these as a function of θ_w and M_∞ .
- What is the net wave drag of this configuration at the design Mach number?

$$(a) \quad C_{p1} = \frac{-2(-\theta_w)}{\sqrt{M^2-1}} = \frac{2\theta_w}{\sqrt{M^2-1}}$$

$$C_{p2} = \frac{2(\theta_w)}{\sqrt{M^2-1}} = \frac{2\theta_w}{\sqrt{M^2-1}}$$

$$C_{p3} = C_{p1} + \frac{2\theta_w}{\sqrt{M^2-1}} = \frac{4\theta_w}{\sqrt{M^2-1}}$$

$$C_{p4} = C_{p3} + \frac{-2\theta_w}{\sqrt{M^2-1}} = \frac{2\theta_w}{\sqrt{M^2-1}}$$

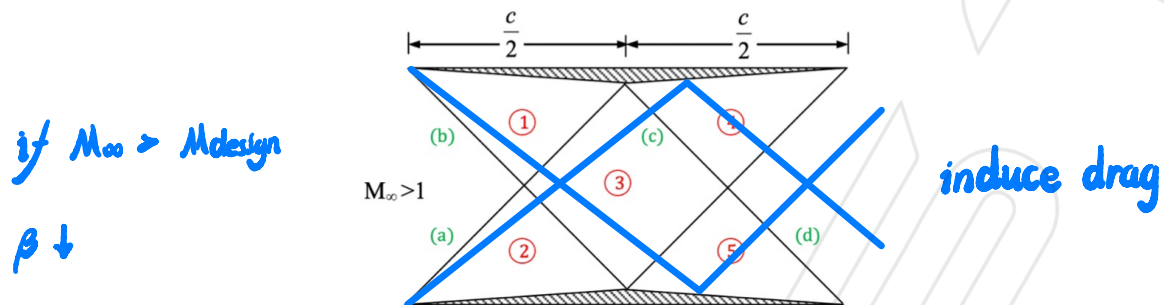
$$C_{p5} = C_{p3} + \frac{2(-\theta_w)}{\sqrt{M^2-1}} = \frac{2\theta_w}{\sqrt{M^2-1}}$$

$$(b) \quad D_{net} = 0$$

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